8-mic array layouts to try

Eight layouts for the XIAO ESP32-S3 (one clock on D1, data on D0, D3, D4, D5), compared for finding where a sound comes from and for steering a beam at the wearer's mouth. Pick a layout, a frequency and a direction to steer toward.

Short version

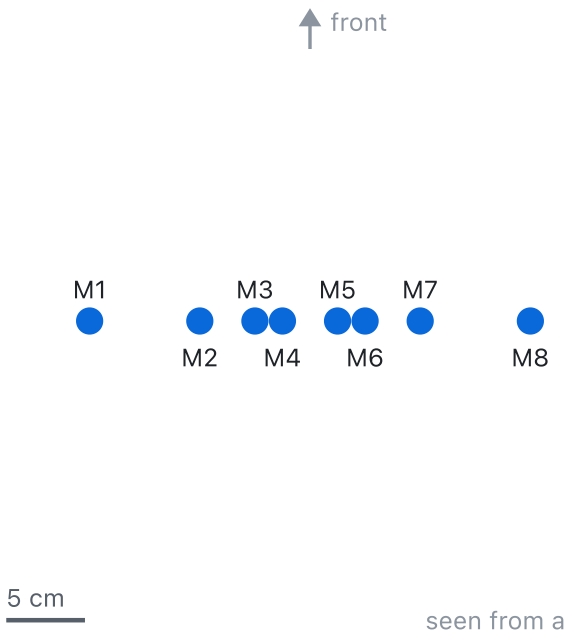
On the bench, add four mics to today's rig as the nested line, then build the 9 cm ring for 360° direction. On the vest, the chest-line variant beats the planned layout for direction finding, while the planned layout's vertical pairs are better for picking up the wearer's voice. Testing both on the mannequin settles it.

Nested line (keeps today's pairs) | Line, 4 cm spacing | Line, 2 cm spacing | Ring, 9 cm across | Ring, 20 cm across | Chest patch, 2 × 4 grid | Vest layout (planned) | Vest, chest line + shoulders + back

Frequency: 500 Hz | 1 kHz | 2 kHz | 3 kHz | 4 kHz | 6 kHz

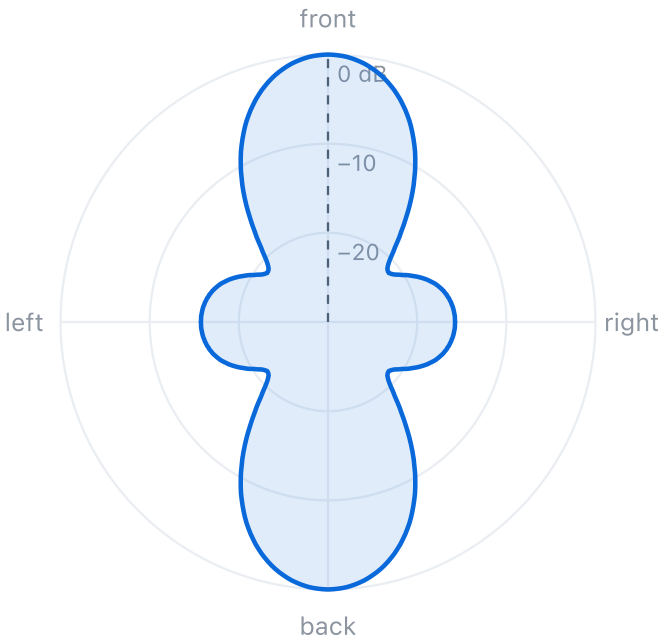
Steer toward: Front | Front-right | Right | Back

Where the mics go



Mics 1–2 on D0, 3–4 on D3, 5–6 on D4, 7–8 on D5 (one mic per pair with SEL to GND, the other to 3V3).

Beam pattern at 2 kHz



Delay-and-sum response vs direction, steered toward the dashed line. Rings at 0, −10, −20 dB; centre is −30 dB.

Nested line (keeps today's pairs)

34°	7.0 dB	all 8 kHz
beam width (−3 dB)	directivity	ghost-free across

Good for
Keeps today's 35 mm and 140 mm pairs and adds four mics, giving spacings from 3.5 to 28 cm. Close mics cover high frequencies and far ones low frequencies, so there are no ghost directions anywhere in the 0–8 kHz band.

Watch out
Can't tell front from back, like every straight line. Uneven spacing raises the side lobes a little.

Where it fits
Easiest next step: four more mics on the same bench rail.

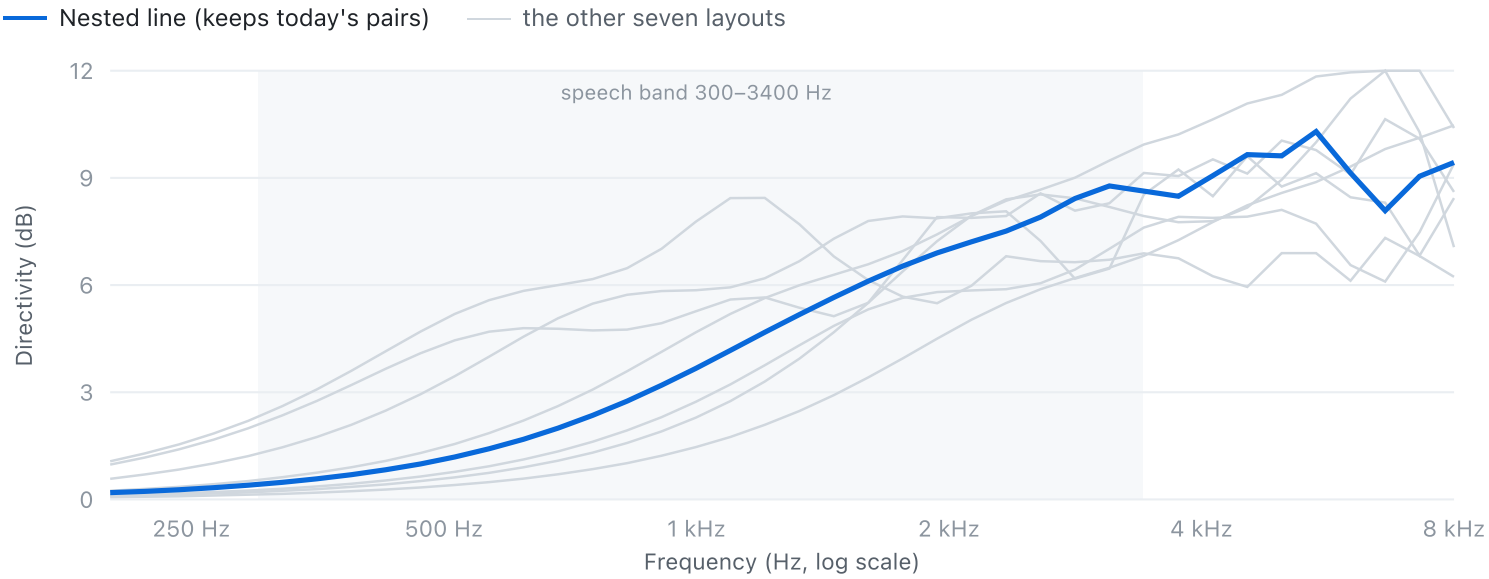
All eight at 2 kHz, steered front

Layout	Span	Beam width	Directivity	Front/back rejection	Ghost-free up to
Nested line (keeps today's pairs)	28 cm	34°	7.0 dB	0 dB (can't tell)	8 kHz (all)
Line, 4 cm spacing	28 cm	28°	7.6 dB	0 dB (can't tell)	4.2 kHz
Line, 2 cm spacing	14 cm	58°	4.7 dB	0 dB (can't tell)	8 kHz (all)
Ring, 9 cm across	9 cm	80°	5.8 dB	9.3 dB	6 kHz
Ring, 20 cm across	20 cm	36°	7.9 dB	4.4 dB	2.9 kHz
Chest patch, 2 × 4 grid	13 cm	58°	7.5 dB	19.6 dB	4.2 kHz
Vest layout (planned)	30 cm	24°	5.6 dB	11.6 dB	600 Hz

Vest, chest line + shoulders + back	28 cm	26°	8.0 dB	11.6 dB	3.8 kHz

Beam width: angle between the −3 dB points (narrower = sharper). Directivity: how much the beam favours its direction over sound from all around (higher = better). Front/back rejection: how much quieter the mirror direction is. Ghost-free: highest frequency before a false direction within 3 dB of the true one appears, worst case over the front half. Model: free-field delay-and-sum, plane waves, c = 343 m/s, horizontal plane only; the body's shadow is not modelled.

Directivity vs frequency, steered front



Directivity of the delay-and-sum beam, 200 Hz to 8 kHz (the board samples at 16 kHz). 9 dB is the most eight mics can give against noise from all around. Same model as the table.

Noise cancellation later: what each method needs from the layout

Method	Layout it needs	Rough gain	Cost on the ESP32-S3
Fixed beam at the mouth (delay-and-sum)	Mics in line with the mouth: vertical chest pairs, or a chest line steered up	Up to 9 dB for 8 mics on uncorrelated noise; less in a reverberant room	Low: one delay and add per mic
Superdirective / MVDR beam	Compact, calibrated cluster (9 cm ring or 2 cm line)	A few dB more than delay-and-sum at low frequencies	Moderate; needs matched mic gains or it amplifies hiss
Adaptive noise canceller with reference mics	Back or shoulder mics that hear the noise but less of the voice (the body blocks about 8 dB)	Large against one loud machine; small in noise from everywhere	Moderate: an adaptive filter per reference mic
Single-channel noise suppression after the beam	Any layout (our noise_compare.py, or ESP-SR's noise suppressor)	Synthetic test: beam + suppression +17.5 dB SNR; real 2-mic take not yet	Low with ESP-SR's WebRTC suppressor

Raw path kept for alarms	Any layout: one mic bypasses the cleanup	Keeps backup alarms and shouts audible	None

What this means for the layout

Put most mics on the upper front chest, in line with the mouth, spaced 4 cm or less so the voice beam works up to about 4 kHz.

Keep two mics on the back. They are the noise references for the adaptive canceller, and they give front/back direction.

These two goals pull in different directions: the 9 cm ring is the best direction finder but a poor voice pickup on the body, so the vest is always a compromise.

Pipeline order

Mics → beam at the mouth → adaptive canceller (back mics as reference) → noise suppressor → radio. A separate raw path feeds the hazard detector so alarms are never filtered out.

Learn the noise while push-to-talk is up; adapt only while the wearer isn't talking.

Suggested test order

Step	Build	Test	Shows
1	Nested line: line up today's pairs on one centre (35 mm pair at ±1.75 cm, 140 mm pair at ±7 cm) and add four mics at ±3.5 cm and ±14 cm	Claps at 1 m from 7 directions across the front; compare direction error with today's 2- and 4-mic results	How much eight mics improve direction over four
2	9 cm ring on a 3D-printed or cardboard disc	Claps from 8 directions all around, including behind	360° direction with front/back: the booth demo
3	Planned vest layout on the mannequin	Same claps, plus talking with factory noise played from a speaker; run noise_compare.py	Body shadow, voice pickup, and noise reduction on the real garment
4	Vest chest-line variant on the mannequin	Same as step 3	Which vest layout to build for the final demo

Eight mics need two more data lines: D4 (GPIO5) and D5 (GPIO6), both slots each. The firmware change from four_mic_stream is small; the USB stream doubles to about 256 kB/s, well within what the board can send.